

bwconvhull

Generate convex hull image from binary image

Syntax

```
CH = bwconvhull(BW)
CH = bwconvhull(BW,method)
CH = bwconvhull(BW,'objects',conn)
```

Description

CH = bwconvhull(BW) computes the convex hull of all objects in BW and returns CH, a binary convex hull image. example

CH = bwconvhull(BW,method) specifies the desired method for computing the convex hull image.

CH = bwconvhull(BW,'objects',conn) specifies the desired connectivity used when defining individual foreground objects.

Examples

[collapse all](#)

▼ Display Binary Convex Hull of Image

Read a grayscale image into the workspace. Convert it into a binary image and calculate the union binary convex hull. Finally, calculate the objects convex hull and display all the images in one figure window.

Try This Example

 [Copy Command](#)

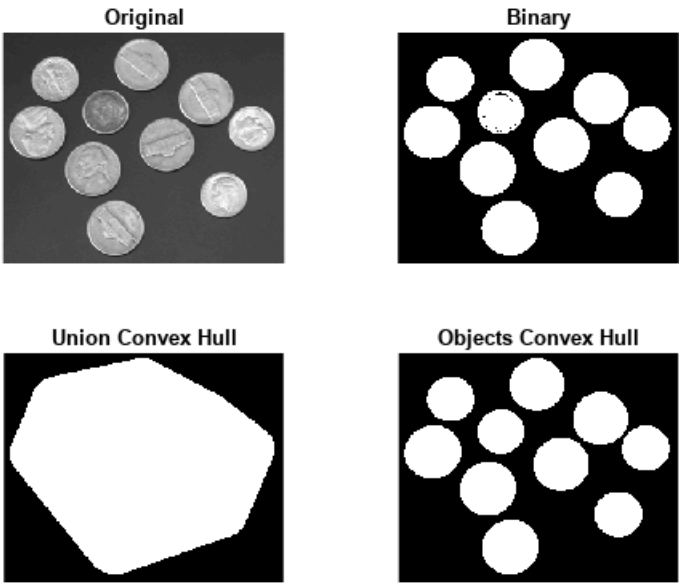
```
subplot(2,2,1);
I = imread('coins.png');
imshow(I);
title('Original');

subplot(2,2,2);
BW = I > 100;
imshow(BW);
title('Binary');

subplot(2,2,3);
CH = bwconvhull(BW);
imshow(CH);
title('Union Convex Hull');

subplot(2,2,4);
CH_objects = bwconvhull(BW,'objects');
imshow(CH_objects);
title('Objects Convex Hull');
```

 Get ▼



Input Arguments

collapse all

▼ **BW — Input binary image**
2-D logical matrix

Input binary image, specified as a 2-D logical matrix.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64 | logical

▼ **method — Method used to compute the convex hull**
'union' (default) | 'objects'

Method used to compute the convex hull, specified as one of the following:

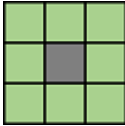
Value	Description
'union'	Compute the convex hull of all foreground objects, treating them as a single object
'objects'	Compute the convex hull of each connected component of BW individually. CH contains the convex hulls of each connected component.

Data Types: char | string

▼ **conn — Pixel connectivity**
8 (default) | 4 | 3-by-3 matrix of 0s and 1s

Pixel connectivity, specified as one of these values The conn parameter is only valid when the method is 'objects'.

Value	Meaning
Two-Dimensional Connectivities	
4	Pixels are connected if their edges touch. Two adjoining pixels are part of the same object if they are both on and are connected along the horizontal or vertical direction.  Current pixel is shown in gray.

Value	Meaning	
8	Pixels are connected if their edges or corners touch. Two adjoining pixels are part of the same object if they are both on and are connected along the horizontal, vertical, or diagonal direction.	 Current pixel is shown in gray.

Connectivity can also be defined in a more general way by specifying a 3-by-3 matrix of 0s and 1s. The 1-valued elements define neighborhood locations relative to the center element of conn. The matrix must be symmetric about its center element.

Data Types: double

Output Arguments

collapse all

✓

CH — Binary mask of the convex hull of all foreground objects in the input image
2-D logical matrix

Binary mask of the convex hull of all foreground objects in the input image, returned as a 2-D logical matrix.

Version History

Introduced in R2011a

See Also

[bwconncomp](#) | [bwlabel](#) | [labelmatrix](#) | [regionprops](#)